

Computing Linear Maps

AICE1004 Linear Algebra

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Representing Linear Maps with Matrices

Matrix representation of a linear map

Definition Suppose that V and W are vector spaces of dimensions n and m with bases B and D , and that $h: V \rightarrow W$ is a linear map. If

$$\text{Rep}_D(h(\vec{\beta}_1)) = \begin{pmatrix} h_{1,1} \\ h_{2,1} \\ \vdots \\ h_{m,1} \end{pmatrix}_D \quad \dots \quad \text{Rep}_D(h(\vec{\beta}_n)) = \begin{pmatrix} h_{1,n} \\ h_{2,n} \\ \vdots \\ h_{m,n} \end{pmatrix}_D$$

then

$$\text{Rep}_{B,D}(h) = \begin{pmatrix} h_{1,1} & h_{1,2} & \dots & h_{1,n} \\ h_{2,1} & h_{2,2} & \dots & h_{2,n} \\ & \vdots & & \\ h_{m,1} & h_{m,2} & \dots & h_{m,n} \end{pmatrix}_{B,D}$$

is the *matrix representation of h with respect to B, D* .

Example Consider projection $\pi: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ onto the x -axis.

$$\begin{pmatrix} a \\ b \end{pmatrix} \mapsto \begin{pmatrix} a \\ 0 \end{pmatrix}$$

Let these be the domain and codomain bases.

$$B = \left\langle \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\rangle \quad D = \left\langle \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \end{pmatrix} \right\rangle$$

Apply the map to each element of the domain basis, and represent the result with respect to the codomain basis

$$\begin{aligned} \begin{pmatrix} 1 \\ 1 \end{pmatrix} &\mapsto \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{so} \quad \text{Rep}_D(\pi(\vec{\beta}_1)) = \begin{pmatrix} -1 \\ 1/2 \end{pmatrix} \\ \begin{pmatrix} -1 \\ 1 \end{pmatrix} &\mapsto \begin{pmatrix} -1 \\ 0 \end{pmatrix} \quad \text{so} \quad \text{Rep}_D(\pi(\vec{\beta}_2)) = \begin{pmatrix} 1 \\ -1/2 \end{pmatrix} \end{aligned}$$

to get the matrix representation of the map.

$$\text{Rep}_{B,D}(\pi) = \begin{pmatrix} -1 & 1 \\ 1/2 & -1/2 \end{pmatrix}$$

Example Again consider projection onto the x -axis

$$\begin{pmatrix} a \\ b \end{pmatrix} \mapsto \begin{pmatrix} a \\ 0 \end{pmatrix}$$

but this time take the input and output bases to be the standard.

$$B = D = \mathcal{E}_2 = \left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\rangle$$

Remembering that with respect to the standard basis each vector represents itself,

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \mapsto \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{so } \text{Rep}_D(\pi(\vec{\beta}_1)) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix} \mapsto \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \text{so } \text{Rep}_D(\pi(\vec{\beta}_2)) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

this is $\text{Rep}_{\mathcal{E}_2, \mathcal{E}_2}(\pi)$.

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

Example We will represent $h: \mathbb{R}^2 \rightarrow \mathbb{R}$

$$\begin{pmatrix} a \\ b \end{pmatrix} \xrightarrow{h} 2a + 3b$$

with respect to $\mathcal{E}_2$ and $\mathcal{E}_1$. First find the effect of h on the domain's basis.

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \mapsto 2 \quad \begin{pmatrix} 0 \\ 1 \end{pmatrix} \mapsto 3$$

Represent those outputs with respect to the codomain's basis.

$$\text{Rep}_{\mathcal{E}_1}(h(\vec{e}_1)) = (2) \quad \text{Rep}_{\mathcal{E}_1}(h(\vec{e}_2)) = (3)$$

(Those column vectors are 1-tall.) This is 1×2 matrix representation.

$$H = \text{Rep}_{\mathcal{E}_2, \mathcal{E}_1}(h) = \begin{pmatrix} 2 & 3 \end{pmatrix}$$

Theorem Assume that V and W are vector spaces of dimensions n and m with bases B and D , and that $h: V \rightarrow W$ is a linear map. If h is represented by

$$\text{Rep}_{B,D}(h) = \begin{pmatrix} h_{1,1} & h_{1,2} & \cdots & h_{1,n} \\ h_{2,1} & h_{2,2} & \cdots & h_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ h_{m,1} & h_{m,2} & \cdots & h_{m,n} \end{pmatrix}_{B,D}$$

and $\vec{v} \in V$ is represented by

$$\text{Rep}_B(\vec{v}) = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix}_B$$

then the representation of the image of $\vec{v}$ is this.

$$\text{Rep}_D(h(\vec{v})) = \begin{pmatrix} h_{1,1}c_1 + h_{1,2}c_2 + \cdots + h_{1,n}c_n \\ h_{2,1}c_1 + h_{2,2}c_2 + \cdots + h_{2,n}c_n \\ \vdots \\ h_{m,1}c_1 + h_{m,2}c_2 + \cdots + h_{m,n}c_n \end{pmatrix}_D$$

Definition The *matrix-vector product* of a $m \times n$ matrix and a $n \times 1$ vector is this.

$$\begin{pmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & & & \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n} \end{pmatrix} \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} = \begin{pmatrix} a_{1,1}c_1 + \cdots + a_{1,n}c_n \\ a_{2,1}c_1 + \cdots + a_{2,n}c_n \\ \vdots \\ a_{m,1}c_1 + \cdots + a_{m,n}c_n \end{pmatrix}$$

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Example We can perform the matrix-vector product operation without reference to maps, or spaces and bases.

$$\begin{pmatrix} 3 & 1 & 2 \\ 0 & -2 & 5 \end{pmatrix} \begin{pmatrix} 4 \\ -1 \\ -3 \end{pmatrix} = \begin{pmatrix} 3 \cdot 4 + 1 \cdot (-1) + 2 \cdot (-3) \\ 0 \cdot 4 - 2 \cdot (-1) + 5 \cdot (-3) \end{pmatrix} = \begin{pmatrix} 5 \\ -13 \end{pmatrix}$$

Example The product of these two is not defined.

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 5 \\ 6 \\ 7 \end{pmatrix}$$

Example Recall the matrix representing projection $\pi: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ onto the x -axis

$$\text{Rep}_{B,D}(\pi) = \begin{pmatrix} -1 & 1 \\ 1/2 & -1/2 \end{pmatrix}$$

with respect to these.

$$B = \left\langle \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\rangle \quad D = \left\langle \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \end{pmatrix} \right\rangle$$

This domain vector

$$\vec{v} = \begin{pmatrix} -1 \\ 5 \end{pmatrix}$$

has this representation with respect to the domain basis.

$$\text{Rep}_B(\vec{v}) = \begin{pmatrix} 2 \\ 3 \end{pmatrix} \quad \text{since} \quad \begin{pmatrix} -1 \\ 5 \end{pmatrix} = 2 \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} + 3 \cdot \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

The matrix-vector product gives the representation of $\pi(\vec{v})$.

$$\begin{pmatrix} -1 & 1 \\ 1/2 & -1/2 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ -1/2 \end{pmatrix}$$

We can check that the prior computation is right. The projection of the vector

$$\pi\left(\begin{pmatrix} -1 \\ 5 \end{pmatrix}\right) = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$

is represented with respect to the codomain basis D as here.

$$\begin{pmatrix} -1 \\ 0 \end{pmatrix} = 1 \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} - (1/2) \cdot \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

Example Recall also that the map $h: \mathbb{R}^2 \rightarrow \mathbb{R}$ with this action

$$\begin{pmatrix} a \\ b \end{pmatrix} \xrightarrow{h} 2a + 3b$$

is represented with respect to the standard bases $\mathcal{E}_2, \mathcal{E}_1$ by a 1×2 matrix.

$$\text{Rep}_{\mathcal{E}_2, \mathcal{E}_1}(h) = \begin{pmatrix} 2 & 3 \end{pmatrix}$$

The domain vector

$$\vec{v} = \begin{pmatrix} -2 \\ 2 \end{pmatrix} \quad \text{Rep}_{\mathcal{E}_2}(\vec{v}) = \begin{pmatrix} -2 \\ 2 \end{pmatrix}$$

has this image.

$$\text{Rep}_{\mathcal{E}_1}(h(\vec{v})) = \begin{pmatrix} 2 & 3 \end{pmatrix} \begin{pmatrix} -2 \\ 2 \end{pmatrix} = (2)_{\mathcal{E}_1}$$

Since this is a representation with respect to the standard basis $\mathcal{E}_1$, meaning that vectors represent themselves, the image is $h(\vec{v}) = 2$.

Any Matrix Represents a Linear Map

The prior subsection shows how to start with a linear map and produce its matrix representation. What about the converse?

Example Fix a matrix

$$H = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

and also fix a domain and codomain, with bases.

$$\mathcal{E}_2 \subset \mathbb{R}^2 \quad \langle 1 - x, 1 + x \rangle \subset \mathcal{P}_1$$

Is there a linear map between the spaces associated with the matrix?

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Is there a linear map between the spaces associated with the matrix?

Consider $h: \mathbb{R}^2 \rightarrow \mathcal{P}_1$ defined by: for any domain vector $\vec{v}$, represent it with respect to the domain basis

$$\vec{v} = c_1 \vec{e}_1 + c_2 \vec{e}_2 \quad \text{Rep}_{\mathcal{E}_2}(\vec{v}) = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

multiply that representation by H

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} c_1 + 2c_2 \\ 3c_1 + 4c_2 \end{pmatrix}$$

and then call $h(\vec{v})$ the codomain vector represented by the result.

$$h(\vec{v}) = (c_1 + 2c_2) \cdot (1 - x) + (3c_1 + 4c_2) \cdot (1 + x)$$

Theorem Any matrix represents a homomorphism between vector spaces of appropriate dimensions, with respect to any pair of bases.

This equation is the key.

$$\text{Rep}_{B,D}(h) \cdot \text{Rep}_B(\vec{v}) = \text{Rep}_D(h(\vec{v})) \quad (*)$$

In this section we will write the domain vector $\vec{v}$ using the entries x, y , etc., and write the codomain vector $h(\vec{v})$ with a, b , etc.

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Example First, the number of rows and columns. This is an instance of (*).

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix}$$

For matrix-vector multiplication to be defined, this 2×3 matrix must multiply a 3-tall input and get a 2-tall output. So, given a matrix H , the dimension of h 's domain is the number of columns in H , and the dimension of h 's codomain is the number of rows.

Rank

Theorem The rank of a matrix equals the rank of any map that it represents.

Example Consider the linear transformation $t: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by

$$\begin{pmatrix} a \\ b \end{pmatrix} \mapsto \begin{pmatrix} 2a - b \\ 2a - b \end{pmatrix}$$

Its range is the line $x = y$ and so the rank of the map is 1.

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We will see two matrices representing this map, the first with respect to these.

$$B = \left\langle \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\rangle, \quad D = \left\langle \begin{pmatrix} 1/2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1/3 \end{pmatrix} \right\rangle$$

This calculation is routine.

$$\text{Rep}_D(t(\vec{\beta}_1)) = \text{Rep}_D\left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 2 \\ 3 \end{pmatrix} \quad \text{Rep}_D\left(\begin{pmatrix} -3 \\ -3 \end{pmatrix}\right) = \begin{pmatrix} -6 \\ -9 \end{pmatrix}$$

The representing matrix has matrix rank 1 since the second row is 3/2 of the first.

$$\text{Rep}_{B,D}(t) = \begin{pmatrix} 2 & -6 \\ 3 & -9 \end{pmatrix}$$

Still considering $t: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by

$$\begin{pmatrix} a \\ b \end{pmatrix} \mapsto \begin{pmatrix} 2a - b \\ 2a - b \end{pmatrix}$$

this time represent it with respect to the standard bases $\mathcal{E}_2, \mathcal{E}_2$.

$$t(\vec{e}_1) = t\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 2 \\ 2 \end{pmatrix} \quad \text{Rep}_{\mathcal{E}_2}\left(\begin{pmatrix} 2 \\ 2 \end{pmatrix}\right) = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

$$t(\vec{e}_2) = t\left(\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} -1 \\ -1 \end{pmatrix} \quad \text{Rep}_{\mathcal{E}_2}\left(\begin{pmatrix} -1 \\ -1 \end{pmatrix}\right) = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$$

(Remember that with respect to the standard basis, a vector represents itself.)

$$\text{Rep}_{\mathcal{E}_2, \mathcal{E}_2}(t) = \begin{pmatrix} 2 & -1 \\ 2 & -1 \end{pmatrix}$$

Gauss's Method on that matrix ends with one nonzero row so it has matrix rank 1, as on the prior slide.

One-to-One, Onto and Singularity

Corollary Let h be a linear map represented by a matrix H . Then h is onto if and only if the rank of H equals the number of its rows, and h is one-to-one if and only if the rank of H equals the number of its columns.

Recall that the number of columns is the dimension of h 's domain, and the number of rows is the dimension of the codomain.

Definition A linear map that is one-to-one and onto is *nonsingular*, otherwise it is *singular*. That is, a linear map is nonsingular if and only if it is an isomorphism.

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Lemma A nonsingular linear map is represented by a square matrix. A square matrix represents nonsingular maps if and only if it is a nonsingular matrix. Thus, a matrix represents isomorphisms if and only if it is square and nonsingular.

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Lemma A nonsingular linear map is represented by a square matrix. A square matrix represents nonsingular maps if and only if it is a nonsingular matrix. Thus, a matrix represents isomorphisms if and only if it is square and nonsingular.

Proof Proof as exercise.

Example This matrix

$$\begin{pmatrix} 0 & 3 \\ -1 & 2 \end{pmatrix}$$

is nonsingular since its two rows form a linearly independent set (we see that by inspection). So any map, with any domain and codomain, and represented by this matrix with respect to any pair of bases, is an isomorphism.

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is nonsingular since its two rows form a linearly independent set (we see that by inspection). So any map, with any domain and codomain, and represented by this matrix with respect to any pair of bases, is an isomorphism.

Example Gauss's method shows that this matrix

$$\begin{pmatrix} 2 & 1 & -2 \\ 3 & 2 & 1 \\ -1 & 0 & 5 \end{pmatrix}$$

is singular so any map that it represents will be a homomorphism that is not an isomorphism.

Computing Range and Null Spaces

For the calculations we use $\mathcal{E}_3 \subseteq \mathbb{R}^3$ for both the domain and codomain.
Find the action of h on the domain's basis

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \mapsto \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \mapsto \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \mapsto \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

and represent those outputs with respect to the codomain's basis.

$$\text{Rep}_{\mathcal{E}_3} \left(\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad \text{Rep}_{\mathcal{E}_3} \left(\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right) = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad \text{Rep}_{\mathcal{E}_3} \left(\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \right) = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

(The standard basis makes for easy calculations.)

$$H = \text{Rep}_{\mathcal{E}_3, \mathcal{E}_3}(h) = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & -1 \end{pmatrix}$$

This is the instance of $\text{Rep}_{B,D}(h) \cdot \text{Rep}_B(\vec{v})$.

$$\begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

To find the range space solve for x , y , and z .

$$\begin{pmatrix} 1 & 0 & 0 & | & a \\ 1 & 0 & 1 & | & b \\ 1 & 0 & -1 & | & c \end{pmatrix} \xrightarrow[-\rho_1 + \rho_3]{-\rho_1 + \rho_2} \begin{pmatrix} 1 & 0 & 0 & | & a \\ 0 & 0 & 1 & | & -a + b \\ 0 & 0 & -1 & | & -a + c \end{pmatrix}$$
$$\xrightarrow{\rho_2 + \rho_3} \begin{pmatrix} 1 & 0 & 0 & | & a \\ 0 & 0 & 1 & | & -a + b \\ 0 & 0 & 0 & | & -2a + b + c \end{pmatrix}$$

This is the instance of $\text{Rep}_{B,D}(h) \cdot \text{Rep}_B(\vec{v})$.

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To find the range space solve for x , y , and z .

$$\begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & -1 \end{pmatrix} \begin{array}{c} a \\ b \\ c \end{array} \xrightarrow[\begin{array}{c} -\rho_1 + \rho_2 \\ -\rho_1 + \rho_3 \end{array}]{\begin{array}{c} -\rho_1 + \rho_2 \\ -\rho_1 + \rho_3 \end{array}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & -1 \end{pmatrix} \begin{array}{c} a \\ -a + b \\ -a + c \end{array}$$

$$\xrightarrow{\rho_2 + \rho_3} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{array}{c} a \\ -a + b \\ -2a + b + c \end{array}$$

The rank is 2. So h is not onto: specifically, the range space $\mathcal{R}(h)$ consists of only those $\vec{w}$'s whose representations satisfy that $0 = -2a + b + c$.

To decide if h is one-to-one we solve the associated homogeneous system.

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & -1 & 0 \end{array}\right) \longrightarrow \longrightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

The solution set

$$\left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mid x = 0 \text{ and } y = 0 \right\} = \left\{ \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \cdot z \mid z \in \mathbb{R} \right\}$$

is nontrivial so h is not one-to-one. Specifically, the null space $\mathcal{N}(h)$ is the set of domain vectors whose representation falls in that one-dimensional set.

Example Let h be the transformation of $\mathbb{R}^2$ represented with respect to the standard basis by this matrix.

$$H = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

This matrix has rank 2 since the second row is not a multiple of the first. This is the instance of equation (*).

$$\text{Rep}_{B,D}(h) \cdot \text{Rep}_B(\vec{v}) = \text{Rep}_D(\vec{w})$$

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix}$$

Do the calculation to solve this linear system for x and y .

$$\begin{aligned} & \left(\begin{array}{cc|c} 1 & 2 & a \\ 3 & 4 & b \end{array} \right) \xrightarrow{-3\rho_1 + \rho_2} \left(\begin{array}{cc|c} 1 & 2 & a \\ 0 & -2 & -3a + b \end{array} \right) \\ & \xrightarrow{-(1/2)\rho_2} \left(\begin{array}{cc|c} 1 & 2 & a \\ 0 & 1 & (3/2)a - (1/2)b \end{array} \right) \xrightarrow{-2\rho_2 + \rho_1} \left(\begin{array}{cc|c} 1 & 0 & -2a + b \\ 0 & 1 & (3/2)a - (1/2)b \end{array} \right) \end{aligned}$$

(Continued from prior slide.)

$$\left(\begin{array}{cc|c} 1 & 2 & a \\ 3 & 4 & b \end{array} \right) \xrightarrow{-3\rho_1 + \rho_2} \cdots \left(\begin{array}{cc|c} 1 & 0 & -2a + b \\ 0 & 1 & (3/2)a - (1/2)b \end{array} \right)$$

The calculation shows that for all codomain vectors

$$\vec{w} = \begin{pmatrix} a \\ b \end{pmatrix}$$

there is an associated domain $\vec{v}$, namely with $x = -2a + b$ and $y = (3/2)a - (1/2)b$. So h is onto, $\mathcal{R}(h) = \mathbb{R}^2$, and the rank of h is as large as it could be, 2.

(Continued from prior slide.)

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Doing the homogeneous system by substituting $a = 0$, $b = 0$ above shows the null space is trivial. So h is one-to-one;

$$\mathcal{N}(h) = \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}$$

The trivial null space has an empty basis, so the nullity is 0.

Example Suppose that a map h between real spaces is represented by this.

$$H = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 3 & 0 \end{pmatrix}$$

The domain has dimension 3 and the codomain has dimension 2 so

$h: \mathbb{R}^3 \rightarrow \mathbb{R}^2$.

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & a \\ 2 & 3 & 0 & b \end{array} \right) \xrightarrow{-2\rho_1 + \rho_2} \xrightarrow{-\rho_2} \xrightarrow{-2\rho_2 + \rho_1} \left(\begin{array}{ccc|c} 1 & 0 & -3 & -3a + 2b \\ 0 & 1 & 2 & 2a - b \end{array} \right)$$

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$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & a \\ 2 & 3 & 0 & b \end{array} \right) \xrightarrow{-2\rho_1 + \rho_2} \xrightarrow{-\rho_2} \xrightarrow{-2\rho_2 + \rho_1} \left(\begin{array}{ccc|c} 1 & 0 & -3 & -3a + 2b \\ 0 & 1 & 2 & 2a - b \end{array} \right)$$

For each a, b there is at least one triple x, y, z so this map is onto;
 $\text{rank}(h) = 2$.

For each a, b there is more than one triple x, y, z so this map is not one-to-one. Said another way, find the null space by substituting $a = 0$, $b = 0$ above to solve the homogeneous system.

$$\mathcal{N}(h) = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mid y + 2z = 0 \text{ and } x - 3z = 0 \right\} = \left\{ \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix} z \mid z \in \mathbb{R} \right\}$$

The nullity is 1.

Leverage

So we can determine things about a map by fixing bases and calculating with the representation.

We close with an example showing how to extend those calculations to the underlying spaces.

Matrix Computation: Sums and Scalar Products

Definition of matrix sum and scalar multiple

Definition The *scalar multiple* of a matrix is the result of entry-by-entry scalar multiplication. The *sum* of two same-sized matrices is their entry-by-entry sum.

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Example Where

$$A = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix} \quad B = \begin{pmatrix} 0 & 0 & 2 \\ 9 & -1/2 & 5 \end{pmatrix} \quad C = \begin{pmatrix} 1 & 0 \\ 8 & -1 \end{pmatrix}$$

Then

$$A + C = \begin{pmatrix} 2 & -1 \\ 10 & 2 \end{pmatrix} \quad 5B = \begin{pmatrix} 0 & 0 & 10 \\ 45 & -5/2 & 25 \end{pmatrix}$$

None of these is defined: $A + B$, $B + A$, $B + C$, $C + B$.

From the definition, they are not defined because the sizes don't match and so the entry-by-entry sum is not possible. But really they are not defined because the underlying function operations are not possible. The fact that A has two columns means that functions represented by A have two-dimensional domains. Functions represented by B have three-dimensional domains. Adding the two functions would be adding apples and oranges.

Theorem Let $h, g: V \rightarrow W$ be linear maps represented with respect to bases B, D by the matrices H and G and let r be a scalar. Then with respect to B, D the map $r \cdot h: V \rightarrow W$ is represented by rH and the map $h + g: V \rightarrow W$ is represented by $H + G$.

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Definition A **zero matrix** has all entries 0. We write $Z_{n \times m}$ or simply Z (another common notation is $0_{n \times m}$ or just 0).

Example The zero matrix is the identity element for matrix addition.

$$\begin{pmatrix} 3 & 1 & 2 \\ 5 & 0 & 9 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 3 & 1 & 2 \\ 5 & 0 & 9 \end{pmatrix}$$

A zero function $Z: V \rightarrow W$ is the identity element for function addition, and the matrix fact accords with the map fact.

Matrix Multiplication

Representing composition

Another function operation, besides scalar multiplication and addition, is composition.

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Another function operation, besides scalar multiplication and addition, is composition.

Lemma The composition of linear maps is linear.

Proof Let $h: V \rightarrow W$ and $g: W \rightarrow U$ be linear. The calculation

$$\begin{aligned} g \circ h (c_1 \cdot \vec{v}_1 + c_2 \cdot \vec{v}_2) &= g(h(c_1 \cdot \vec{v}_1 + c_2 \cdot \vec{v}_2)) = g(c_1 \cdot h(\vec{v}_1) + c_2 \cdot h(\vec{v}_2)) \\ &= c_1 \cdot g(h(\vec{v}_1)) + c_2 \cdot g(h(\vec{v}_2)) = c_1 \cdot (g \circ h)(\vec{v}_1) + c_2 \cdot (g \circ h)(\vec{v}_2) \end{aligned}$$

shows that $g \circ h: V \rightarrow U$ preserves linear combinations, and so is linear.

QED

Example Consider two linear functions $h: V \rightarrow W$ and $g: W \rightarrow X$ represented as here.

$$\text{Rep}_{B,C}(h) = \begin{pmatrix} 3 & 1 \\ 2 & 5 \\ 4 & 6 \end{pmatrix} \quad \text{Rep}_{C,D}(g) = \begin{pmatrix} 8 & 7 & 11 \\ 9 & 10 & 12 \end{pmatrix}$$

We will do an explanatory computation, to see how these two representations combine to give the representation of the composition $g \circ h: V \rightarrow X$.

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We will do an explanatory computation, to see how these two representations combine to give the representation of the composition $g \circ h: V \rightarrow X$.

We start with the action of h on $\vec{v} \in V$.

$$\begin{aligned} \text{Rep}_C(h(\vec{v})) &= \text{Rep}_{B,C}(h) \cdot \text{Rep}_B(\vec{v}) \\ &= \begin{pmatrix} 3 & 1 \\ 2 & 5 \\ 4 & 6 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 3v_1 + v_2 \\ 2v_1 + 5v_2 \\ 4v_1 + 6v_2 \end{pmatrix} \end{aligned}$$

Next, to that apply g .

$$\begin{aligned}\text{Rep}_{C,D}(g) \cdot \text{Rep}_C(h(\vec{v})) &= \begin{pmatrix} 8 & 7 & 11 \\ 9 & 10 & 12 \end{pmatrix} \begin{pmatrix} 3v_1 + v_2 \\ 2v_1 + 5v_2 \\ 4v_1 + 6v_2 \end{pmatrix} \\ &= \begin{pmatrix} 8(3v_1 + v_2) + 7(2v_1 + 5v_2) + 11(4v_1 + 6v_2) \\ 9(3v_1 + v_2) + 10(2v_1 + 5v_2) + 12(4v_1 + 6v_2) \end{pmatrix}\end{aligned}$$

Gather terms.

$$= \begin{pmatrix} (8 \cdot 3 + 7 \cdot 2 + 11 \cdot 4)v_1 + (8 \cdot 1 + 7 \cdot 5 + 11 \cdot 6)v_2 \\ (9 \cdot 3 + 10 \cdot 2 + 12 \cdot 4)v_1 + (9 \cdot 1 + 10 \cdot 5 + 12 \cdot 6)v_2 \end{pmatrix}$$

Rewrite as a matrix-vector multiplication.

$$= \begin{pmatrix} 8 \cdot 3 + 7 \cdot 2 + 11 \cdot 4 & 8 \cdot 1 + 7 \cdot 5 + 11 \cdot 6 \\ 9 \cdot 3 + 10 \cdot 2 + 12 \cdot 4 & 9 \cdot 1 + 10 \cdot 5 + 12 \cdot 6 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$$

So here is how the two starting matrices combine.

$$\begin{pmatrix} 8 & 7 & 11 \\ 9 & 10 & 12 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ 2 & 5 \\ 4 & 6 \end{pmatrix} = \begin{pmatrix} 8 \cdot 3 + 7 \cdot 2 + 11 \cdot 4 & 8 \cdot 1 + 7 \cdot 5 + 11 \cdot 6 \\ 9 \cdot 3 + 10 \cdot 2 + 12 \cdot 4 & 9 \cdot 1 + 10 \cdot 5 + 12 \cdot 6 \end{pmatrix}$$

Definition of matrix multiplication

Definition The **matrix-multiplicative product** of the $m \times r$ matrix G and the $r \times n$ matrix H is the $m \times n$ matrix P , where

$$p_{i,j} = g_{i,1}h_{1,j} + g_{i,2}h_{2,j} + \cdots + g_{i,r}h_{r,j}$$

so that the i, j -th entry of the product is the dot product of the i -th row of the first matrix with the j -th column of the second.

$$GH = \begin{pmatrix} & \vdots & \\ g_{i,1} & g_{i,2} & \cdots & g_{i,r} \\ & \vdots & \end{pmatrix} \begin{pmatrix} \cdots & h_{1,j} & \cdots \\ & h_{2,j} & \\ & \vdots & \\ & h_{r,j} & \end{pmatrix} = \begin{pmatrix} & \vdots & \\ \cdots & p_{i,j} & \cdots \\ & \vdots & \end{pmatrix}$$

Example

$$\begin{pmatrix} 3 & 1 & 6 \\ 2 & 5 & 9 \end{pmatrix} \begin{pmatrix} 2 & 0 & 4 \\ 1 & -3 & 5 \\ 4 & 2 & 7 \end{pmatrix} = \begin{pmatrix} 31 & 9 & 59 \\ 45 & 3 & 96 \end{pmatrix}$$

Example This product

$$\begin{pmatrix} 1 & 3 & -1 \\ 0 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix} \begin{pmatrix} 5 & 7 & 1 \\ 2 & 2 & 0 \end{pmatrix}$$

is not defined because the number of columns on the left must equal the number of rows on the right.

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Example Square matrices of the same size have a defined product.

$$\begin{pmatrix} 1 & 3 & -1 \\ 0 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix} \begin{pmatrix} 5 & 7 & 1 \\ 2 & 2 & 0 \\ 1 & -1 & 2 \end{pmatrix} = \begin{pmatrix} 10 & 14 & -1 \\ 0 & 0 & 0 \\ 10 & 14 & 2 \end{pmatrix}$$

This reflects the fact that we can compose two functions from a space to itself $g, h: V \rightarrow V$.

Matrix multiplication represents composition

2.7 *Theorem* A composition of linear maps is represented by the matrix product of the representatives.

Arrow diagrams

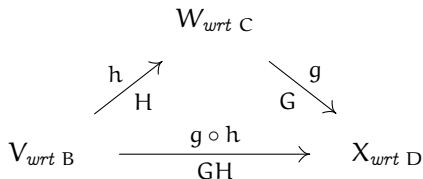
This pictures the relationship between maps and matrices.

$$\begin{array}{ccc} & W_{\text{wrt } C} & \\ \nearrow h & & \nwarrow g \\ & & \\ V_{\text{wrt } B} & \xrightarrow[\text{GH}]{g \circ h} & X_{\text{wrt } D} \end{array}$$

The diagram illustrates the relationship between maps and matrices. It shows three vector spaces: $V_{\text{wrt } B}$ on the left, $W_{\text{wrt } C}$ at the top, and $X_{\text{wrt } D}$ on the right. A map h (represented by matrix H) maps $V_{\text{wrt } B}$ to $W_{\text{wrt } C}$. A map g (represented by matrix G) maps $W_{\text{wrt } C}$ to $X_{\text{wrt } D}$. The composition map $g \circ h$ (represented by matrix GH) maps $V_{\text{wrt } B}$ directly to $X_{\text{wrt } D}$.

Arrow diagrams

This pictures the relationship between maps and matrices.



Above the arrows, the maps show that the two ways of going from V to X , straight over via the composition or else in two steps by way of W , have the same effect

$$\vec{v} \xrightarrow{g \circ h} g(h(\vec{v})) \quad \vec{v} \xrightarrow{h} h(\vec{v}) \xrightarrow{g} g(h(\vec{v}))$$

(this is just the definition of composition). Below the arrows, the matrices indicate that multiplying GH into the column vector $\text{Rep}_B(\vec{v})$ has the same effect as multiplying the column vector first by H and then multiplying the result by G .

$$\text{Rep}_{B,D}(g \circ h) = GH \quad \text{Rep}_{C,D}(g) \text{Rep}_{B,C}(h) = GH$$

Order, dimensions, and sizes

An important observation about the order in which we write these things: in writing the composition $g \circ h$, the function g is written first, that is, leftmost, but it is applied second.

$$\vec{v} \xrightarrow{h} h(\vec{v}) \xrightarrow{g} g(h(\vec{v}))$$

That order carries over to matrices: $g \circ h$ is represented by GH .

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An important observation about the order in which we write these things: in writing the composition $g \circ h$, the function g is written first, that is, leftmost, but it is applied second.

$$\vec{v} \xrightarrow{h} h(\vec{v}) \xrightarrow{g} g(h(\vec{v}))$$

That order carries over to matrices: $g \circ h$ is represented by GH .

Also consider the dimensions of the spaces.

$$\text{dimension } n \text{ space} \xrightarrow{h} \text{dimension } r \text{ space} \xrightarrow{g} \text{dimension } m \text{ space}$$

Briefly, $m \times r$ times $r \times n$ equals $m \times n$, as here.

$$\begin{array}{ccc} 2 \times 3 & 3 \times 4 & = \quad 2 \times 4 \\ \begin{pmatrix} 2 & 1 & 4 \\ -1 & 0 & 3 \end{pmatrix} \begin{pmatrix} 3 & 0 & 2 & 1 \\ 5 & 0 & 0 & 2 \\ 1 & -1 & 4 & 7 \end{pmatrix} & = & \begin{pmatrix} 15 & -4 & 20 & 32 \\ 0 & -3 & 10 & 20 \end{pmatrix} \end{array}$$

Matrix multiplication is not commutative

Function composition is in general not a commutative operation— $\cos(\sqrt{x})$ is different than $\sqrt{\cos(x)}$. This holds even in the special case of composition of linear functions.

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Example Changing the order in which we multiply these matrices

$$\begin{pmatrix} 3 & 3 \\ 0 & 4 \end{pmatrix} \begin{pmatrix} -2 & 6 \\ 6 & 5 \end{pmatrix} = \begin{pmatrix} 12 & 33 \\ 24 & 20 \end{pmatrix}$$

changes the result.

$$\begin{pmatrix} -2 & 6 \\ 6 & 5 \end{pmatrix} \begin{pmatrix} 3 & 3 \\ 0 & 4 \end{pmatrix} = \begin{pmatrix} -6 & 18 \\ 18 & 38 \end{pmatrix}$$

Example The product of these two is defined in one order and not defined in the other.

$$\begin{pmatrix} 3 & 4 \\ 0 & 2 \end{pmatrix} \quad \begin{pmatrix} 8 & 12 & 0 \\ -4 & 0 & 1/2 \end{pmatrix}$$

Although the matrix operation of multiplication does not have the property of being commutative, it does have some nice algebraic properties.

Theorem If F , G , and H are matrices, and the matrix products are defined, then the product is associative $(FG)H = F(GH)$ and distributes over matrix addition $F(G + H) = FG + FH$ and $(G + H)F = GF + HF$.

Proof Associativity holds because matrix multiplication represents function composition, which is associative: the maps $(f \circ g) \circ h$ and $f \circ (g \circ h)$ are equal as both send $\vec{v}$ to $f(g(h(\vec{v})))$.

Distributivity is similar. For instance, the first one goes $f \circ (g + h)(\vec{v}) = f((g + h)(\vec{v})) = f(g(\vec{v}) + h(\vec{v})) = f(g(\vec{v})) + f(h(\vec{v})) = f \circ g(\vec{v}) + f \circ h(\vec{v})$ (the third equality uses the linearity of f). Right-distributivity goes the same way. QED

Mechanics of Matrix Multiplication

Combinatorics of multiplication

The striking thing about matrix multiplication is the way rows and columns combine.

Lemma In a product of two matrices G and H , the columns of GH are formed by taking G times the columns of H

$$G \cdot \begin{pmatrix} \vdots & & \vdots \\ \vec{h}_1 & \cdots & \vec{h}_n \\ \vdots & & \vdots \end{pmatrix} = \begin{pmatrix} \vdots & & \vdots \\ G \cdot \vec{h}_1 & \cdots & G \cdot \vec{h}_n \\ \vdots & & \vdots \end{pmatrix}$$

and the rows of GH are formed by taking the rows of G times H

$$\begin{pmatrix} \cdots & \vec{g}_1 & \cdots \\ & \vdots & \\ \cdots & \vec{g}_r & \cdots \end{pmatrix} \cdot H = \begin{pmatrix} \cdots & \vec{g}_1 \cdot H & \cdots \\ & \vdots & \\ \cdots & \vec{g}_r \cdot H & \cdots \end{pmatrix}$$

Definition A matrix with all 0's except for a 1 in the i, j entry is an i, j *unit matrix* (or *matrix unit*).

Example The 2, 1 unit 2×3 matrix multiplies from the left

$$\begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 2 & 1 & 3 \\ 5 & 6 & 4 \\ 7 & 8 & 9 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 2 & 1 & 3 \end{pmatrix}$$

to copy row 1 of the multiplicand into row 2 of the result.

Example From the right the 2, 1 unit 2×3 matrix

$$\begin{pmatrix} 3 & 4 \\ 6 & 5 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 4 & 0 & 0 \\ 5 & 0 & 0 \end{pmatrix}$$

copies column 2 of the first matrix into column 1 of the result.

Example Rescaling the unit matrix rescales the result.

$$\begin{pmatrix} 3 & 4 \\ 6 & 5 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 3 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 12 & 0 & 0 \\ 15 & 0 & 0 \end{pmatrix}$$

Definition The *main diagonal* (or *principal diagonal* or simply *diagonal*) of a square matrix goes from the upper left to the lower right.

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Definition An *identity matrix* is square and every entry is 0 except for 1's in the main diagonal.

$$I_{n \times n} = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ & \vdots & & \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

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Taking the product with an identity matrix returns the multiplicand.

Example Multiplication by an identity from the left

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & 2 \\ -1 & 5 \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ -1 & 5 \end{pmatrix}$$

or from the right leaves the matrix unchanged.

$$\begin{pmatrix} 3 & 2 \\ -1 & 5 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ -1 & 5 \end{pmatrix}$$

Definition A *diagonal matrix* is square and has 0's off the main diagonal.

$$\begin{pmatrix} a_{1,1} & 0 & \dots & 0 \\ 0 & a_{2,2} & \dots & 0 \\ & \vdots & & \\ 0 & 0 & \dots & a_{n,n} \end{pmatrix}$$

Example Multiplication from the left by a diagonal matrix rescales the rows.

$$\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 3 & 2 \\ -1 & 5 \end{pmatrix} = \begin{pmatrix} 6 & 4 \\ -3 & 15 \end{pmatrix}$$

From the right it rescales the columns.

$$\begin{pmatrix} 3 & 2 \\ -1 & 5 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 6 & 6 \\ -2 & 15 \end{pmatrix}$$

Definition A *permutation matrix* is square and is all 0's except for a single 1 in each row and column.

Example Multiplication by a permutation matrix from the left will swap rows.

$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} = \begin{pmatrix} 4 & 5 & 6 \\ 1 & 2 & 3 \\ 7 & 8 & 9 \end{pmatrix}$$

From the right it swaps columns.

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 3 \\ 5 & 4 & 6 \\ 8 & 7 & 9 \end{pmatrix}$$

Definition The *elementary reduction matrices* (or just *elementary matrices*) result from applying a single Gaussian operation to an identity matrix.

- 1) $I \xrightarrow{k\rho_i} M_i(k)$ for $k \neq 0$
- 2) $I \xrightarrow{\rho_i \leftrightarrow \rho_j} P_{i,j}$ for $i \neq j$
- 3) $I \xrightarrow{k\rho_i + \rho_j} C_{i,j}(k)$ for $i \neq j$

Example Multiplying on the left by the 3×3 matrix $M_2(1/2)$ has the effect of the row operation $(1/2)\rho_2$.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 3 & 0 \\ 0 & 2 & 5 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 3 & 0 \\ 0 & 1 & 5/2 \\ 0 & 0 & 0 \end{pmatrix}$$

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Left multiplication by $C_{1,3}(-2)$ performs the row operation $-2\rho_1 + \rho_3$.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 3 & 0 \\ 0 & 2 & 5 \\ 2 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 3 & 0 \\ 0 & 2 & 5 \\ 0 & -5 & 1 \end{pmatrix}$$

Lemma Matrix multiplication can do Gaussian reduction.

- 1) If $H \xrightarrow{k\rho_i} G$ then $M_i(k)H = G$.
- 2) If $H \xrightarrow{\rho_i \leftrightarrow \rho_j} G$ then $P_{i,j}H = G$.
- 3) If $H \xrightarrow{k\rho_i + \rho_j} G$ then $C_{i,j}(k)H = G$.

Corollary For any matrix H there are elementary reduction matrices $R_1, \dots, R_r$ such that $R_r \cdot R_{r-1} \cdots R_1 \cdot H$ is in reduced echelon form.

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Corollary For any matrix H there are elementary reduction matrices $R_1, \dots, R_r$ such that $R_r \cdot R_{r-1} \cdots R_1 \cdot H$ is in reduced echelon form.

Example You can do this Gauss-Jordan reduction

$$\begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix} \xrightarrow{-4\rho_1 + \rho_2} \begin{pmatrix} 1 & 2 \\ 0 & -5 \end{pmatrix} \xrightarrow{-(1/5)\rho_2} \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \xrightarrow{-2\rho_2 + \rho_1} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

as a product; note that the order is right-to-left.

$$\begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1/5 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -4 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$C_{2,1}(-2) M_2(-1/5) C_{1,2}(-4) \cdot H = I$$

Example You can bring this augmented matrix to echelon form with matrix multiplication.

$$\left(\begin{array}{ccc|c} 1 & -1 & 2 & 4 \\ 2 & -2 & -1 & 6 \\ 0 & 3 & 1 & 5 \end{array}\right)$$

First perform $-2\rho_1 + \rho_2$ via left multiplication by $C_{1,2}(-2)$.

$$\left(\begin{array}{ccc} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right) \left(\begin{array}{ccc|c} 1 & -1 & 2 & 4 \\ 2 & -2 & -1 & 6 \\ 0 & 3 & 1 & 5 \end{array}\right) = \left(\begin{array}{ccc|c} 1 & -1 & 2 & 4 \\ 0 & 0 & -5 & -2 \\ 0 & 3 & 1 & 5 \end{array}\right)$$

Swap rows 2 and 3 with $P_{2,3}$

$$\left(\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{array}\right) \left(\begin{array}{ccc|c} 1 & -1 & 2 & 4 \\ 0 & 0 & -5 & -2 \\ 0 & 3 & 1 & 5 \end{array}\right) = \left(\begin{array}{ccc|c} 1 & -1 & 2 & 4 \\ 0 & 3 & 1 & 5 \\ 0 & 0 & -5 & -2 \end{array}\right)$$

Here is the full equation.

$$\left(\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{array}\right) \left(\begin{array}{ccc} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right) \left(\begin{array}{ccc|c} 1 & -1 & 2 & 4 \\ 2 & -2 & -1 & 6 \\ 0 & 3 & 1 & 5 \end{array}\right) = \left(\begin{array}{ccc|c} 1 & -1 & 2 & 4 \\ 0 & 3 & 1 & 5 \\ 0 & 0 & -5 & -2 \end{array}\right)$$

Inverses

Function inverses

Our goal is, where a linear $\mathfrak{h}$ has an inverse, to find the relationship between the matrices $\text{Rep}_{B,D}(\mathfrak{h})$ and $\text{Rep}_{D,B}(\mathfrak{h}^{-1})$.

Function inverses

Our goal is, where a linear h has an inverse, to find the relationship between the matrices $\text{Rep}_{B,D}(h)$ and $\text{Rep}_{D,B}(h^{-1})$.

We first recall some things about inverses. Where $\pi: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ is the projection map and $\iota: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is the embedding

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \xrightarrow{\pi} \begin{pmatrix} x \\ y \end{pmatrix} \quad \begin{pmatrix} x \\ y \end{pmatrix} \xrightarrow{\iota} \begin{pmatrix} x \\ y \\ 0 \end{pmatrix}$$

then the composition $\pi \circ \iota$ is the identity map $\pi \circ \iota = \text{id}$ on $\mathbb{R}^2$.

$$\begin{pmatrix} x \\ y \end{pmatrix} \xrightarrow{\iota} \begin{pmatrix} x \\ y \\ 0 \end{pmatrix} \xrightarrow{\pi} \begin{pmatrix} x \\ y \end{pmatrix}$$

We say that ι is a *right inverse* of π or, what is the same thing, that π is a *left inverse* of ι .

However, composition in the other order $\iota \circ \pi$ doesn't give the identity map—here is a vector that is not sent to itself under $\iota \circ \pi$.

$$\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \xrightarrow{\pi} \begin{pmatrix} 0 \\ 0 \end{pmatrix} \xrightarrow{\iota} \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

However, composition in the other order $\iota \circ \pi$ doesn't give the identity map—here is a vector that is not sent to itself under $\iota \circ \pi$.

$$\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \xrightarrow{\pi} \begin{pmatrix} 0 \\ 0 \end{pmatrix} \xrightarrow{\iota} \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

In fact, π has no left inverse at all. For, if f were to be a left inverse of π then we would have

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \xrightarrow{\pi} \begin{pmatrix} x \\ y \end{pmatrix} \xrightarrow{f} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

for all of the infinitely many z 's. But a function f cannot send a single argument $\begin{pmatrix} x \\ y \end{pmatrix}$ to more than one value.

So a function can have a right inverse but no left inverse, or a left inverse but no right inverse. A function can also fail to have an inverse on either side; one example is the zero transformation on $\mathbb{R}^2$.

Definition of matrix inverse

Definition A matrix G is a *left inverse matrix* of the matrix H if GH is the identity matrix. It is a *right inverse* if HG is the identity. A matrix H with a two-sided inverse is an *invertible matrix*. That two-sided inverse is denoted H^{-1} .

Definition of matrix inverse

Definition A matrix G is a *left inverse matrix* of the matrix H if GH is the identity matrix. It is a *right inverse* if HG is the identity. A matrix H with a two-sided inverse is an *invertible matrix*. That two-sided inverse is denoted H^{-1} .

Example This matrix

$$H = \begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix}$$

has a two-sided inverse.

$$H^{-1} = \begin{pmatrix} 3 & -5 \\ -1 & 2 \end{pmatrix}$$

To check that, we multiply them in both orders. Here is one; the other is just as easy.

$$\begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 3 & -5 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Solving linear systems using matrix inverse

One advantage of knowing a matrix inverse is that it makes solving a linear system easy and quick. To solve

$$2x + 5y = -3$$

$$x + 3y = 10$$

rewrite as a matrix equation

$$\begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -3 \\ 10 \end{pmatrix}$$

and multiply both sides (from the left) by the matrix inverse.

$$\begin{pmatrix} 3 & -5 \\ -1 & 2 \end{pmatrix} \cdot \begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 & -5 \\ -1 & 2 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ 10 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -59 \\ 23 \end{pmatrix}$$

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This specializes the arrow diagram for composition to the case of inverses.

$$\begin{array}{ccc}
 & W_{wrt\ C} & \\
 \nearrow h & & \nwarrow h^{-1} \\
 & H & H^{-1} \\
 V_{wrt\ B} & \xrightarrow[\text{I}]{\text{id}} & V_{wrt\ B}
 \end{array}$$

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 \end{array}$$

Theorem A matrix is invertible if and only if it is nonsingular.

Proof (For both results.) Given a matrix H , fix spaces of appropriate dimension for the domain and codomain and fix bases for these spaces. With respect to these bases, H represents a map h . The statements are true about the map and therefore they are true about the matrix. QED

Lemma A product of invertible matrices is invertible: if G and H are invertible and GH is defined then GH is invertible and $(GH)^{-1} = H^{-1}G^{-1}$.

Note The order changes for each individual matrix.

Lemma A matrix H is invertible if and only if it can be written as the product of elementary reduction matrices. We can compute the inverse by applying to the identity matrix the same row steps, in the same order, that Gauss-Jordan reduce H .

Derivation of matrix inverse by Gauss-Jordan Reduction

Example This matrix is nonsingular and so is invertible.

$$A = \begin{pmatrix} 1 & 3 & 1 \\ 2 & 0 & -1 \\ 1 & 2 & 0 \end{pmatrix}$$

We write the matrix A next to the 3×3 identity and as we Gauss-Jordan reduce the matrix on the left, we apply those operations also on the right.

$$\begin{aligned} \left(\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 2 & 0 & -1 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 0 & 1 \end{array} \right) & \xrightarrow[\begin{smallmatrix} -2\rho_1+\rho_2 \\ -\rho_1+\rho_3 \end{smallmatrix}]{} \left(\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 0 & -6 & -3 & -2 & 1 & 0 \\ 0 & -1 & -1 & -1 & 0 & 1 \end{array} \right) \\ & \xrightarrow{-1/6\rho_2+\rho_3} \left(\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 0 & -6 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1/2 & -2/3 & -1/6 & 1 \end{array} \right) \end{aligned}$$

The left-hand side is in echelon form. We continue with the second half of Gauss-Jordan reduction on the next slide.

$$\xrightarrow[-2\rho_3]{-1/6\rho_2} \left(\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1/2 & 1/3 & -1/6 & 0 \\ 0 & 0 & 1 & 4/3 & 1/3 & -2 \end{array} \right)$$

$$\xrightarrow[-(1/2)\rho_3+\rho_2]{-\rho_3+\rho_1} \left(\begin{array}{ccc|ccc} 1 & 3 & 0 & -1/3 & -1/3 & 2 \\ 0 & 1 & 0 & -1/3 & -1/3 & 1 \\ 0 & 0 & 1 & 4/3 & 1/3 & -2 \end{array} \right)$$

$$\xrightarrow{-3\rho_2+\rho_1} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 2/3 & 2/3 & -1 \\ 0 & 1 & 0 & -1/3 & -1/3 & 1 \\ 0 & 0 & 1 & 4/3 & 1/3 & -2 \end{array} \right)$$

This is the inverse.

$$A^{-1} = \begin{pmatrix} 2/3 & 2/3 & -1 \\ -1/3 & -1/3 & 1 \\ 4/3 & 1/3 & -2 \end{pmatrix}$$

Finding the inverse of a matrix A is a lot of work but as we noted earlier, once we have it then solving linear systems $A\vec{x} = \vec{b}$ is easy.

Example The linear system

$$\begin{array}{rcl} x + 3y + z & = & 2 \\ 2x & & -z = 12 \\ x + 2y & = & 4 \end{array}$$

is this matrix equation.

$$\begin{pmatrix} 1 & 3 & 1 \\ 2 & 0 & -1 \\ 1 & 2 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 12 \\ 4 \end{pmatrix}$$

Solve it by multiplying both sides from the left by the inverse that we found earlier.

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2/3 & 2/3 & -1 \\ -1/3 & -1/3 & 1 \\ 4/3 & 1/3 & -2 \end{pmatrix} \begin{pmatrix} 2 \\ 12 \\ 4 \end{pmatrix} = \begin{pmatrix} 16/3 \\ -2/3 \\ -4/3 \end{pmatrix}$$

We sometimes want to repeatedly solve systems with the same left side but different right sides. This system equals the one on the prior slide but for one number on the right.

$$\begin{array}{rcl} x + 3y + z & = & 1 \\ 2x & - z & = 12 \\ x + 2y & = & 4 \end{array}$$

The solution is this.

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2/3 & 2/3 & -1 \\ -1/3 & -1/3 & 1 \\ 4/3 & 1/3 & -2 \end{pmatrix} \begin{pmatrix} 1 \\ 12 \\ 4 \end{pmatrix} = \begin{pmatrix} 14/3 \\ -1/3 \\ -8/3 \end{pmatrix}$$

The inverse of a 2×2 matrix

Corollary The inverse for a 2×2 matrix exists and equals

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

if and only if $ad - bc \neq 0$.

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Example

$$\begin{pmatrix} 2 & 4 \\ -1 & 1 \end{pmatrix}^{-1} = \frac{1}{6} \begin{pmatrix} 1 & -4 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 1/6 & -2/3 \\ 1/6 & 1/3 \end{pmatrix}$$

Change of Basis: Changing representations of vectors

Coordinates vary with the basis

Consider this vector $\vec{v} \in \mathbb{R}^3$ and bases for the space.

$$\vec{v} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad \mathcal{E}_3 = \left\langle \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\rangle \quad B = \left\langle \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\rangle$$

With respect to the different bases, the coordinates of $\vec{v}$ are different.

$$\text{Rep}_{\mathcal{E}_3}(\vec{v}) = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad \text{Rep}_B(\vec{v}) = \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}$$

In this section we will see how to convert the representation of a vector with respect to a first basis to its representation with respect to a second.

Change of basis matrix

Think of translating from $\text{Rep}_B(\vec{v})$ to $\text{Rep}_D(\vec{v})$ as holding the vector constant. This is the arrow diagram.

$$\begin{array}{c} V_{\text{wrt } B} \\ \text{id} \downarrow \\ V_{\text{wrt } D} \end{array}$$

(This diagram is vertical to fit with the ones in the next subsection.)

Change of basis matrix

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(This diagram is vertical to fit with the ones in the next subsection.)

Definition The *change of basis matrix* for bases $B, D \subset V$ is the representation of the identity map $\text{id}: V \rightarrow V$ with respect to those bases.

$$\text{Rep}_{B,D}(\text{id}) = \begin{pmatrix} \vdots & & \vdots \\ \text{Rep}_D(\vec{\beta}_1) & \cdots & \text{Rep}_D(\vec{\beta}_n) \\ \vdots & & \vdots \end{pmatrix}$$

Lemma To convert from the representation of a vector $\vec{v}$ with respect to B to its representation with respect to D use the change of basis matrix.

$$\text{Rep}_{B,D}(\text{id}) \text{Rep}_B(\vec{v}) = \text{Rep}_D(\vec{v})$$

Conversely, if left-multiplication by a matrix changes bases $M \cdot \text{Rep}_B(\vec{v}) = \text{Rep}_D(\vec{v})$ then M is a change of basis matrix.

Example To change a representation of a member of $\mathcal{P}_2$ from being with respect to $B = \langle 1, 1 + x, 1 + x + x^2 \rangle$ to being with respect to $D = \langle x^2 - 1, x, x^2 + 1 \rangle$, compute $\text{Rep}_{B,D}(\text{id})$. The identity map acting on the elements of B has no effect. Represent those elements with respect to D .

$$\text{Rep}_D(1) = \begin{pmatrix} -1/2 \\ 0 \\ 1/2 \end{pmatrix} \quad \text{Rep}_D(1+x) = \begin{pmatrix} -1/2 \\ 1 \\ 1/2 \end{pmatrix} \quad \text{Rep}_D(1+x+x^2) = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

The change of basis matrix is the concatenation of those.

$$\text{Rep}_{B,D}(\text{id}) = \begin{pmatrix} -1/2 & -1/2 & 0 \\ 0 & 1 & 1 \\ 1/2 & 1/2 & 1 \end{pmatrix}$$

For example, we can translate the representation of $\vec{v} = 2 - x + 3x^2$.

$$\text{Rep}_B(\vec{v}) = \begin{pmatrix} 3 \\ -4 \\ 3 \end{pmatrix} \quad \text{Rep}_D(\vec{v}) = \begin{pmatrix} 1/2 \\ -1 \\ 5/2 \end{pmatrix} = \begin{pmatrix} -1/2 & -1/2 & 0 \\ 0 & 1 & 1 \\ 1/2 & 1/2 & 1 \end{pmatrix} \begin{pmatrix} 3 \\ -4 \\ 3 \end{pmatrix}$$

Lemma A matrix changes bases if and only if it is nonsingular.

Example Recall that any nonsingular matrix M decomposes into a product of elementary reduction matrices $M = R_1 \cdots R_r$. We can show that M changes bases by showing that each R_i changes bases. Consider this 3×3 case.

$$C_{1,3}(-4) \cdot \text{Rep}_B(\vec{v}) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{pmatrix} \begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix} = \begin{pmatrix} r_1 \\ r_2 \\ -4r_1 + r_3 \end{pmatrix}$$

Where $B = \langle \vec{\beta}_1, \vec{\beta}_2, \vec{\beta}_3 \rangle$, the right side represents the same vector $\vec{v}$ with respect to $\hat{B} = \langle \vec{\beta}_1 + 4\vec{\beta}_3, \vec{\beta}_2, \vec{\beta}_3 \rangle$.

$$r_1 \cdot (\vec{\beta}_1 + 4\vec{\beta}_3) + r_2 \cdot \vec{\beta}_2 + (-4r_1 + r_3) \cdot \vec{\beta}_3 = r_1 \vec{\beta}_1 + r_2 \vec{\beta}_2 + r_3 \vec{\beta}_3$$

Verifying that $\hat{B}$ is a basis is routine.

Changing map representations

The natural next step is to see how to convert $\text{Rep}_{B,D}(h)$ to $\text{Rep}_{\hat{B},\hat{D}}(h)$. Here is the arrow diagram.

$$\begin{array}{ccc}
 V_{\text{wrt } B} & \xrightarrow[\quad H \quad]{\quad h \quad} & W_{\text{wrt } D} \\
 \text{id} \downarrow & & \text{id} \downarrow \\
 V_{\text{wrt } \hat{B}} & \xrightarrow[\quad \hat{H} \quad]{\quad h \quad} & W_{\text{wrt } \hat{D}}
 \end{array}$$

To move from the lower-left to the lower-right we can either go straight over, or else up to V_B then over to W_D and then down. So we can calculate $\hat{H} = \text{Rep}_{\hat{B},\hat{D}}(h)$ either by directly using $\hat{B}$ and $\hat{D}$, or else by first changing bases with $\text{Rep}_{\hat{B},B}(\text{id})$ then multiplying by $H = \text{Rep}_{B,D}(h)$ and then changing bases with $\text{Rep}_{D,\hat{D}}(\text{id})$.

Theorem To convert from the matrix H representing a map h with respect to B, D to the matrix $\hat{H}$ representing it with respect to $\hat{B}, \hat{D}$ use this formula.

$$\hat{H} = \text{Rep}_{D,\hat{D}}(\text{id}) \cdot H \cdot \text{Rep}_{\hat{B},B}(\text{id}) \quad (*)$$

Proof This is evident from the diagram.

QED

Example Consider the derivative map $d/dx: \mathcal{P}_2 \rightarrow \mathcal{P}_2$ as well as $B = \langle 1, 1+x, 1+x+x^2 \rangle$, $D = \langle 1+x^2, x, 1-x^2 \rangle$ and $\hat{B} = \langle 1, x, x^2 \rangle$, $\hat{D} = \langle 1+x, x+x^2, 1+x^2 \rangle$.

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We can find H and $\hat{H}$ using the methods we have already seen.

$$\text{Rep}_{B,D}(d/dx) = \begin{pmatrix} 0 & 1/2 & 1/2 \\ 0 & 0 & 2 \\ 0 & 1/2 & 1/2 \end{pmatrix} \quad \text{Rep}_{\hat{B},\hat{D}}(d/dx) = \begin{pmatrix} 0 & 1/2 & 1 \\ 0 & -1/2 & 1 \\ 0 & 1/2 & -1 \end{pmatrix}$$

These do the base changes.

$$\text{Rep}_{\hat{B},B}(\text{id}) = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{Rep}_{D,\hat{D}}(\text{id}) = \begin{pmatrix} 0 & 1/2 & 1 \\ 0 & 1/2 & -1 \\ 1 & -1/2 & 0 \end{pmatrix}$$

We can check this case of equation (*) by multiplying through.

$$\begin{aligned} \text{Rep}_{\hat{B},\hat{D}}(d/dx) &= \text{Rep}_{D,\hat{D}}(\text{id}) \cdot \text{Rep}_{B,D}(d/dx) \cdot \text{Rep}_{\hat{B},B}(\text{id}) \\ &= \begin{pmatrix} 0 & 1/2 & 1 \\ 0 & 1/2 & -1 \\ 1 & -1/2 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1/2 & 1/2 \\ 0 & 0 & 2 \\ 0 & 1/2 & 1/2 \end{pmatrix} \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \end{aligned}$$

With respect to the standard basis real vectors represent themselves, so the matrix representing moving up is easy.

$$\text{Rep}_{B, \mathcal{E}_2}(\text{id}) = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

The matrix for moving down is the inverse of the prior one.

$$\text{Rep}_{\mathcal{E}_2, D}(\text{id}) = (\text{Rep}_{D, \mathcal{E}_2}(\text{id}))^{-1} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}^{-1} = \frac{-1}{2} \cdot \begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix}$$

Thus we have this.

$$\begin{aligned} \text{Rep}_{B, D}(t_{\pi/6}) &= \frac{-1}{2} \cdot \begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} \sqrt{3}/2 & -1/2 \\ 1/2 & \sqrt{3}/2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \\ &= \begin{pmatrix} \sqrt{3}/2 & 1/2 \\ -1/2 & \sqrt{3}/2 \end{pmatrix} \end{aligned}$$

Definition Same-sized matrices H and $\hat{H}$ are *matrix equivalent* if there are nonsingular matrices P and Q such that $\hat{H} = PHQ$.

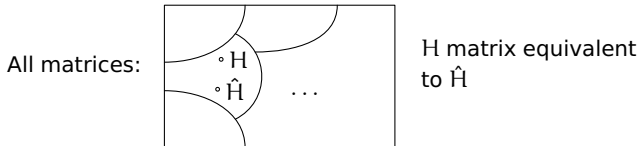
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Corollary Matrix equivalent matrices represent the same map, with respect to appropriate pairs of bases.

Matrix equivalence is an equivalence relation. Thus it partitions the set of matrices into matrix equivalence classes.



Canonical form for matrix equivalence

Theorem Any $m \times n$ matrix of rank k is matrix equivalent to the $m \times n$ matrix that is all zeros except that the first k diagonal entries are ones.

$$\begin{pmatrix} 1 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 & 0 & \dots & 0 \\ & \vdots & & & & & \\ 0 & 0 & \dots & 1 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 & 0 & \dots & 0 \\ & \vdots & & & & & \\ 0 & 0 & \dots & 0 & 0 & \dots & 0 \end{pmatrix}$$

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This is a *block partial-identity* form.

$$\left(\begin{array}{c|c} I & Z \\ \hline Z & Z \end{array} \right)$$

For an example, recall first that any nonsingular matrix M can be factored into a product $M = R_1 \cdots R_r$ where each R_t is an elementary reduction matrix $C_{i,j}(k)$, or $M_i(k)$, or $P_{i,j}$.

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Recall also that from the left matrices operate on rows

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 3 & 0 & -3 \end{pmatrix}$$

while from the right they act on columns: this does $-4 \cdot \text{col}_3 + \text{col}_1$.

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -4 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -11 & 2 & 3 \\ -20 & 5 & 6 \\ -29 & 8 & 9 \end{pmatrix}$$

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Example This matrix has rank 2.

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

We will find P and Q to get the 3×3 canonical matrix of rank 2.

Row operations produce echelon form.

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \xrightarrow[-7\rho_1+\rho_3]{-4\rho_1+\rho_2} \xrightarrow{-2\rho_2+\rho_3} \xrightarrow{-(1/3)\rho_2} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

Multiplying the reduction matrices gives P; note the right to left order.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1/3 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -7 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ = \begin{pmatrix} 1 & 0 & 0 \\ 4/3 & -1/3 & 0 \\ 1 & -2 & 1 \end{pmatrix}$$

So far we have this.

$$PHQ = \begin{pmatrix} 1 & 0 & 0 \\ 4/3 & -1/3 & 0 \\ 1 & -2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} Q = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix} Q$$

Column operations produce the block partial identity.

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{-2\text{col}_2 + \text{col}_3} \xrightarrow{-2\text{col}_1 + \text{col}_2} \xrightarrow{\text{col}_1 + \text{col}_3} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Combine the reduction matrices to get Q. In contrast with the construction of P, here they come left to right.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -2 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix}$$

In sum, we have this equation.

$$PHQ = \begin{pmatrix} 1 & 0 & 0 \\ 4/3 & -1/3 & 0 \\ 1 & -2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} 1 & -2 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Action of a canonical form matrix

This kind of matrix is has an easy to understand effect.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ 0 \end{pmatrix}$$

It is a projection—the map $t: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ represented with respect to the standard bases by this block partial identity matrix is projection from three space to the xy plane.

Matrix equivalence is characterized by rank

Corollary Matrix equivalence classes are characterized by rank: two same-sized matrices are matrix equivalent if and only if they have the same rank.

Matrix equivalence is characterized by rank

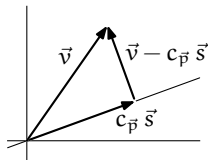
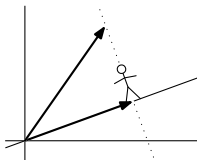
Corollary Matrix equivalence classes are characterized by rank: two same-sized matrices are matrix equivalent if and only if they have the same rank.

Proof Two same-sized matrices with the same rank are equivalent to the same block partial-identity matrix. QED

Projection (Optional)

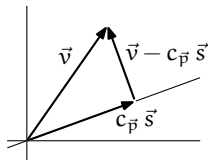
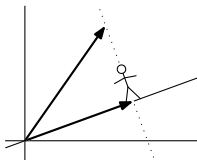
Project a vector into a line

This shows a figure walking out on the line to a point $\vec{p}$ such that the tip of $\vec{v}$ is directly above them, where “above” does not mean parallel to the y-axis but instead means orthogonal to the line.



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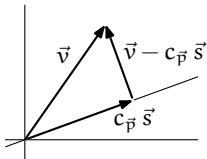
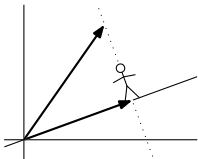
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Since the line is the span of some vector $\ell = \{c \cdot \vec{s} \mid c \in \mathbb{R}\}$, we have a coefficient $c_{\vec{p}}$ with the property that $\vec{v} - c_{\vec{p}}\vec{s}$ is orthogonal to $c_{\vec{p}}\vec{s}$.

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To solve for this coefficient, observe that because $\vec{v} - c_{\vec{p}}\vec{s}$ is orthogonal to a scalar multiple of $\vec{s}$, it must be orthogonal to $\vec{s}$ itself. Then $(\vec{v} - c_{\vec{p}}\vec{s}) \cdot \vec{s} = 0$ gives that $c_{\vec{p}} = \vec{v} \cdot \vec{s} / \vec{s} \cdot \vec{s}$.

Definition The *orthogonal projection of $\vec{v}$ into the line spanned by a nonzero $\vec{s}$* is this vector.

$$\text{proj}_{[\vec{s}]}(\vec{v}) = \frac{\vec{v} \cdot \vec{s}}{\vec{s} \cdot \vec{s}} \cdot \vec{s}$$

Example The projection of this $\mathbb{R}^3$ vector into the line

$$\vec{v} = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \quad L = \{c \cdot \vec{s} = c \cdot \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} \mid c \in \mathbb{R}\}$$

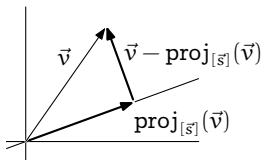
is this vector.

$$\frac{\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}}{\begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}} \cdot \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1/3 \\ 1/6 \\ 1/6 \end{pmatrix}$$

Gram-Schmidt Orthogonalization

Mutually orthogonal vectors

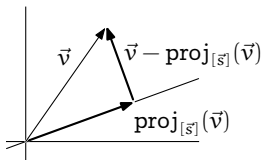
The prior subsection suggests that projecting a vector $\vec{v}$ into the line spanned by $\vec{s}$ decomposes $\vec{v}$ into two parts, a part with the line and a part orthogonal to that.



$$\vec{v} = \text{proj}_{[\vec{s}]}(\vec{v}) + (\vec{v} - \text{proj}_{[\vec{s}]}(\vec{v}))$$

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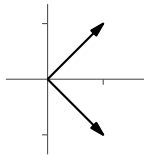
Definition Vectors $\vec{v}_1, \dots, \vec{v}_k \in \mathbb{R}^n$ are *mutually orthogonal* when any two are orthogonal: if $i \neq j$ then the dot product $\vec{v}_i \cdot \vec{v}_j$ is zero.

Example The vectors of the standard basis $\mathcal{E}_3 \subset \mathbb{R}^3$ are mutually orthogonal.

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Example These two vectors in $\mathbb{R}^2$ are mutually orthogonal.

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$



Theorem If the vectors in a set $\{\vec{v}_1, \dots, \vec{v}_k\} \subset \mathbb{R}^n$ are mutually orthogonal and nonzero then that set is linearly independent.

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Corollary In a k dimensional vector space, if the vectors in a size k set are mutually orthogonal and nonzero then that set is a basis for the space.

Definition An *orthogonal basis* for a vector space is a basis of mutually orthogonal vectors.

Theorem If $\langle \vec{\beta}_1, \dots, \vec{\beta}_k \rangle$ is a basis for a subspace of $\mathbb{R}^n$ then the vectors

$$\vec{\kappa}_1 = \vec{\beta}_1$$

$$\vec{\kappa}_2 = \vec{\beta}_2 - \text{proj}_{[\vec{\kappa}_1]}(\vec{\beta}_2)$$

$$\vec{\kappa}_3 = \vec{\beta}_3 - \text{proj}_{[\vec{\kappa}_1]}(\vec{\beta}_3) - \text{proj}_{[\vec{\kappa}_2]}(\vec{\beta}_3)$$

$$\vdots$$

$$\vec{\kappa}_k = \vec{\beta}_k - \text{proj}_{[\vec{\kappa}_1]}(\vec{\beta}_k) - \dots - \text{proj}_{[\vec{\kappa}_{k-1}]}(\vec{\beta}_k)$$

form an orthogonal basis for the same subspace.

Example This basis for $\mathbb{R}^2$

$$B = \left\langle \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \end{pmatrix} \right\rangle$$

does not have orthogonal vectors. To derive from it a basis $K = \langle \vec{\kappa}_1, \vec{\kappa}_2 \rangle$ that is orthogonal, start by taking the first vector unchanged.

$$\vec{\kappa}_1 = \vec{\beta}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

For $\vec{\kappa}_2$ take the part of $\vec{\beta}_2$ that does not lie with $\vec{\kappa}_1$.

$$\vec{\kappa}_2 = \vec{\beta}_2 - \text{proj}_{[\vec{\kappa}_1]}(\vec{\beta}_2) = \begin{pmatrix} 1 \\ 3 \end{pmatrix} - \frac{\begin{pmatrix} 1 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix}}{\begin{pmatrix} 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix}} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -2/5 \\ 1/5 \end{pmatrix}$$

Note that $\vec{\kappa}_1$ and $\vec{\kappa}_2$ are indeed orthogonal.

Example This is a basis for $\mathbb{R}^3$.

$$B = \left\langle \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \\ -1 \end{pmatrix} \right\rangle$$

Start the orthogonal basis with $\vec{\beta}_1$.

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As in the prior slide, the next step is $\vec{\kappa}_2 = \vec{\beta}_2 - \text{proj}_{[\vec{\kappa}_1]}(\vec{\beta}_2)$.

$$\begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} - \frac{\begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}}{\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -3/2 \\ 3/2 \\ 0 \end{pmatrix}$$

The third step is $\vec{\kappa}_3 = \vec{\beta}_3 - \text{proj}_{[\vec{\kappa}_1]}(\vec{\beta}_3) - \text{proj}_{[\vec{\kappa}_2]}(\vec{\beta}_3)$.

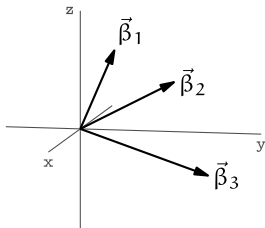
$$\begin{pmatrix} 0 \\ 3 \\ -1 \end{pmatrix} - \frac{\begin{pmatrix} 0 \\ 3 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}}{\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}} \cdot \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} - \frac{\begin{pmatrix} 0 \\ 3 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} -3/2 \\ 3/2 \\ 0 \end{pmatrix}}{\begin{pmatrix} -3/2 \\ 3/2 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} -3/2 \\ 3/2 \\ 0 \end{pmatrix}} \cdot \begin{pmatrix} -3/2 \\ 3/2 \\ 0 \end{pmatrix} = \begin{pmatrix} 4/3 \\ 4/3 \\ -4/3 \end{pmatrix}$$

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The members of B are at odd angles but the members of K are mutually orthogonal.

$$B = \left\langle \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \\ -1 \end{pmatrix} \right\rangle$$

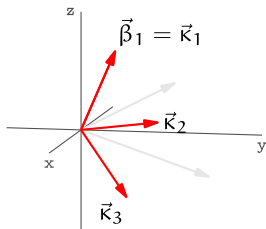


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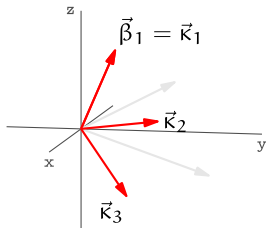


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We could go on to make this basis even more like $\mathcal{E}_3$ by normalizing all of its members to have length 1, making an *orthonormal* basis.